Glutathione S-transferase activity assessment

Material:

1. Sample
2. Potassium phosphate buffer
3. EDTA
4. NADPH
5. Milli-Q water
6. GSH
7. 1-chloro-2,4-dinitrobenzene (CDNB)

Procedures:

1. Homogenize the sample in a sonicator (Branson Sonifier Model 250; three 4-s cycles) in potassium phosphate buffer 100 mM and EDTA 50 mM, pH 7.4, and centrifuge the homogenate at 10.000 rpm and 4 ^{o}C for 10 min.
2. Add 30 µL of the sample to 240 µL of potassium phosphate buffer solution (100 mL of monobasic potassium phosphate buffer, 100 mL of dibasic phosphate buffer; pH 7.5) and 318 µL of milli-Q water. Homogenize.
3. After homogenization, add 6 µL of GSH 100 mM and 6 µL of 1-chloro-2,4-dinitrobenzene (CDNB) 100 mM.
4. Place the reagents directly into the optical cells.
5. Read the absorbance in a spectrophotometer (Hitachi, model U-2001, Hitachinaka City, Japan) at 340 nm, for 120 s, at 30-s intervals.
6. The enzymatic activity is determined according to the CDNB extinction coefficient.